Inference and Proofs

COMP2711 Discrete Mathematical Tools

Rosen, Sections 1.6–1.8

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SECTION 1

Arguments and validity

A proof is a checkable certificate

MUST UNDERSTAND **Proof**

premises
accepted starting statements

A proof is a checkable certificate

MUST UNDERSTAND **Proof**

premises
accepted starting statements

→

inference rules
truth-preserving steps

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conclusion
the claimed result

A proof is a checkable certificate

MUST UNDERSTAND **Proof**

premises
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→

inference rules
truth-preserving steps

→

conclusion
the claimed result

A proof records **where we start**, **why each move is allowed**, and **the exact claim reached**.

Premises and conclusion

$$\begin{array}{c} P_1 \\ P_2 \\ \vdots \\ P_n \\ \hline \therefore C \end{array}$$

Above the line: the **premises**.

Below the line: the **conclusion** C.

The question is whether the premises **force** C.

The line records each statement's **role**; it does not guarantee that the argument is correct.

Valid arguments

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No possible case has all premises **true** and the conclusion **false**.

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$$\boxed{(P_1 \wedge \cdots \wedge P_n) \to C \text{ is a tautology.}}$$

$$\text{all premises true} \xrightarrow{\textbf{forbidden}} \text{conclusion false}$$

Valid arguments

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No possible case has all premises **true** and the conclusion **false**.

$$\boxed{(P_1 \wedge \cdots \wedge P_n) \to C \text{ is a tautology.}}$$

Validity concerns the **connection**, not whether a premise happens to be true.

Countermodels

MUST UNDERSTAND **Countermodel**

Test the argument

$$P \to Q, \qquad Q \qquad \therefore \qquad P.$$

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$$P \to Q, \qquad Q \qquad \therefore \qquad P.$$

To make the conclusion false, choose $P = \mathrm{F}$.
Choose $Q = \mathrm{T}$. Then

$$P \to Q = \mathrm{T}, \qquad Q = \mathrm{T}, \qquad P = \mathrm{F}.$$

Countermodels

MUST UNDERSTAND **Countermodel**

Test the argument

$$P \to Q, \qquad Q \qquad \therefore \qquad P.$$

To make the conclusion false, choose $P = \mathrm{F}$.
Choose $Q = \mathrm{T}$. Then

$$P \to Q = \mathrm{T}, \qquad Q = \mathrm{T}, \qquad P = \mathrm{F}.$$

One true-premises/false-conclusion case proves the argument **invalid**.

SECTION 2

Propositional inference

Rules of inference

MUST UNDERSTAND **Rule of inference**

A reusable argument form that **preserves truth**.

known
statement

Rules of inference

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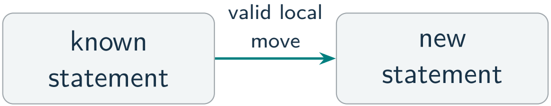
A reusable argument form that **preserves truth**.



Rules of inference

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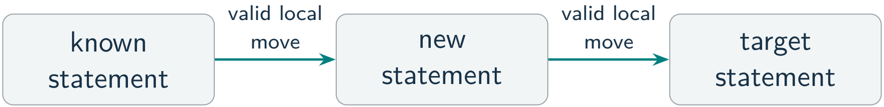

Rules of inference

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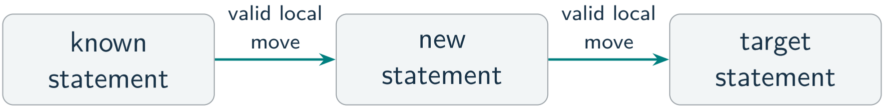

A long proof is auditable because each non-obvious line cites its local reason.

Implication rules

Rule	Form	Job
Modus ponens	$P,\ P \to Q\ \therefore Q$	direct implication

Implication rules

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Modus ponens	$P,\ P \to Q\ \therefore Q$	direct implication
Modus tollens	$\neg Q,\ P \to Q\ \therefore \neg P$	contrapositive

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Hypothetical syllogism	$P \to Q,\ Q \to R\ \therefore P \to R$	chain two implications

Implication rules

Rule	Form	Job
Modus ponens	$P,\ P \to Q\ \therefore Q$	direct implication
Modus tollens	$\neg Q,\ P \to Q\ \therefore \neg P$	contrapositive
Hypothetical syllogism	$P \to Q,\ Q \to R\ \therefore P \to R$	chain two implications

Modus ponens in words: If it rains, the ceremony moves indoors. It is raining, so the ceremony moves indoors.

Conjunction and disjunction rules

Rule	Form	Job
Simplification	$P \wedge Q \ \therefore P$	use one known part
Conjunction	$P, \ Q \ \therefore P \wedge Q$	combine two known facts
Addition	$P \ \therefore P \vee Q$	add an alternative
Disjunctive syllogism	$P \vee Q, \ \neg P \ \therefore Q$	eliminate an alternative

Name the move so that the next reader can check it locally.

Resolution is case analysis

$$P \vee Q, \qquad \neg P \vee R \qquad \therefore \qquad Q \vee R.$$

Resolution is case analysis

$$P \lor Q, \qquad \neg P \lor R \qquad \therefore \qquad Q \lor R.$$

If P is false

The first premise forces Q.

Resolution is case analysis

$$P \vee Q, \qquad \neg P \vee R \qquad \therefore \qquad Q \vee R.$$

If P is false

The first premise forces Q.

If P is true

The second premise forces R.

Resolution is case analysis

$$P \lor Q, \qquad \neg P \lor R \qquad \therefore \qquad Q \lor R.$$

If P is false
The first premise forces Q.

If P is true
The second premise forces R.

Either way, $Q \lor R$ holds. Resolution is just doing case-by-case reasoning.

Bus-delay argument

Let

$$D = \text{"the bus is delayed,"} \qquad C = \text{"the student misses the connection,"}$$
$$L = \text{"the student arrives late."}$$

Travel rule	$D \to C$	If the bus is delayed, the student misses the connection.
Travel rule	$C \to L$	If the student misses the connection, the student arrives late.
Observed premise	$\neg L$	The student did not arrive late.

Is the bus delayed? $\boxed{D = \mathrm{T}}$ or $\boxed{D = \mathrm{F}}$?

Bus-delay solution

Step	Statement	Reason
1	$D \to C$	given travel rule
2	$C \to L$	given travel rule
3	$\neg L$	observed premise

Bus-delay solution

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1	$D \to C$	given travel rule
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4	$\neg C$	modus tollens, 2 and 3

Bus-delay solution

Step	Statement	Reason
1	$D \to C$	given travel rule
2	$C \to L$	given travel rule
3	$\neg L$	observed premise
4	$\neg C$	modus tollens, 2 and 3
5	$\neg D$	modus tollens, 1 and 4

Bus-delay solution

Step	Statement	Reason
1	$D \to C$	given travel rule
2	$C \to L$	given travel rule
3	$\neg L$	observed premise
4	$\neg C$	modus tollens, 2 and 3
5	$\neg D$	modus tollens, 1 and 4

Being on time rules out the missed connection, then the delayed bus.

$D = \mathrm{F}$: the bus was not delayed.

Propositional proof practice

Show that

$$P \to Q, \qquad Q \to R, \qquad P \wedge S \qquad \therefore \qquad R \wedge S.$$

Step	Statement	Reason
1	$P \to Q$	premise
2	$Q \to R$	premise
3	$P \wedge S$	premise

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1	$P \to Q$	premise
2	$Q \to R$	premise
3	$P \wedge S$	premise
4	P	simplification, 3
5	S	simplification, 3

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5	S	simplification, 3
6	Q	modus ponens, 1 and 4

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1	$P \to Q$	premise
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3	$P \wedge S$	premise
4	P	simplification, 3
5	S	simplification, 3
6	Q	modus ponens, 1 and 4
7	R	modus ponens, 2 and 6

Propositional proof practice

Show that

$$P \to Q, \qquad Q \to R, \qquad P \land S \qquad \therefore \qquad R \land S.$$

Step	Statement	Reason
1	$P \to Q$	premise
2	$Q \to R$	premise
3	$P \land S$	premise
4	P	simplification, 3
5	S	simplification, 3
6	Q	modus ponens, 1 and 4
7	R	modus ponens, 2 and 6
8	$R \land S$	conjunction, 7 and 5

Each line supplies exactly what the next rule needs.

Invalid implication directions

$P =$ "the vehicle is a car," $\qquad Q =$ "the vehicle has wheels."

Given rule: $P \to Q$ If the vehicle is a car, then it has wheels.

Affirming the consequent

$$Q \quad \not\Rightarrow \quad P$$

"It has wheels, so it is a car."

Denying the antecedent

$$\neg P \quad \not\Rightarrow \quad \neg Q$$

"It is not a car, so it has no wheels."

Invalid implication directions

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"It has wheels, so it is a car."
Not necessarily: it could be a bicycle.

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Invalid implication directions

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“It is not a car, so it has no wheels.”
Not necessarily: a bicycle is not a car and has wheels.

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"It is not a car, so it has no wheels."
Not necessarily: a bicycle is not a car and has wheels.

The bicycle is a **countermodel**: the premises are true and each claimed conclusion is false.

SECTION 3

Quantified inference

Instantiate or generalize

MUST UNDERSTAND **Quantified inference**

Instantiate

quantified statement $\longrightarrow$ one object

$$\forall x \, P(x) \;\therefore\; P(a)$$

Remove a quantifier.

Instantiate or generalize

MUST UNDERSTAND **Quantified inference**

Instantiate

quantified statement $\longrightarrow$ one object

$$\forall x\, P(x) \;\therefore\; P(a)$$

Remove a quantifier.

Generalize

one object $\longrightarrow$ quantified statement

$$P(a) \;\therefore\; \exists x\, P(x)$$

Add a quantifier.

Instantiate or generalize

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quantified statement $\longrightarrow$ one object

$$\forall x\, P(x) \;\therefore\; P(a)$$

Remove a quantifier.

Generalize

one object $\longrightarrow$ quantified statement

$$P(a) \;\therefore\; \exists x\, P(x)$$

Add a quantifier.

These are controlled inference rules, not automatic equivalences.

Instantiation rules

Universal instantiation

$$\forall x\, P(x) \quad \therefore P(a)$$

Every object has P, so any particular a has P.

Instantiation rules

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$$\forall x\, P(x) \;\; \therefore P(a)$$

Every object has P, so any particular a has P.

Existential instantiation

$$\exists x\, P(x) \;\; \therefore P(c)$$

Introduce a **fresh name** c for one promised object.

Instantiation rules

Universal instantiation

$$\forall x\, P(x) \;\; \therefore P(a)$$

Every object has P, so any particular a has P.

Existential instantiation

$$\exists x\, P(x) \;\; \therefore P(c)$$

Introduce a **fresh name** c for one promised object.

The existential move is a controlled proof convention, not a logical equivalence.

Generalization rules

Universal generalization

$$P(a) \;\therefore \forall x\, P(x)$$

a was arbitrary; no premise or temporary assumption restricts it.

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Existential generalization

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One verified object establishes existence.

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a was arbitrary; no premise or temporary assumption restricts it.

Existential generalization

$$P(a) \;\therefore \exists x\, P(x)$$

One verified object establishes existence.

The side condition is part of the rule.

Quantified argument

Domain: all students.

$$\begin{aligned} R(x) &: \text{“}x \text{ registered for the exam,”} \\ A(x) &: \text{“}x \text{ was assigned a seat,”} \\ L(x) &: \text{“}x \text{ arrived late.”} \end{aligned}$$

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Premises:

$$\forall x\,(R(x) \to A(x)), \qquad \exists x\,(R(x) \wedge L(x)).$$

Quantified argument

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Premises:

$$\forall x\,(R(x) \to A(x)), \qquad \exists x\,(R(x) \wedge L(x)).$$

Goal:

$$\boxed{\exists x\,(A(x) \wedge L(x))}$$

Some student who was assigned a seat arrived late.

Quantified argument solution

Step	Statement	Reason
1	$\exists x\,(R(x) \wedge L(x))$	premise

Quantified argument solution

Step	Statement	Reason
1	$\exists x\,(R(x) \wedge L(x))$	premise
2	$R(c) \wedge L(c)$	existential instantiation; c fresh
3	$R(c)$	simplification, 2
4	$L(c)$	simplification, 2

Quantified argument solution

Step	Statement	Reason
1	$\exists x\,(R(x) \wedge L(x))$	premise
2	$R(c) \wedge L(c)$	existential instantiation; c fresh
3	$R(c)$	simplification, 2
4	$L(c)$	simplification, 2
5	$\forall x\,(R(x) \to A(x))$	premise
6	$R(c) \to A(c)$	universal instantiation, 5
7	$A(c)$	modus ponens, 3 and 6

Quantified argument solution

Step	Statement	Reason
1	$\exists x\,(R(x) \wedge L(x))$	premise
2	$R(c) \wedge L(c)$	existential instantiation; c fresh
3	$R(c)$	simplification, 2
4	$L(c)$	simplification, 2
5	$\forall x\,(R(x) \rightarrow A(x))$	premise
6	$R(c) \rightarrow A(c)$	universal instantiation, 5
7	$A(c)$	modus ponens, 3 and 6
8	$A(c) \wedge L(c)$	conjunction, 7 and 4

Quantified argument solution

Step	Statement	Reason
1	$\exists x\,(R(x) \land L(x))$	premise
2	$R(c) \land L(c)$	existential instantiation; c fresh
3	$R(c)$	simplification, 2
4	$L(c)$	simplification, 2
5	$\forall x\,(R(x) \to A(x))$	premise
6	$R(c) \to A(c)$	universal instantiation, 5
7	$A(c)$	modus ponens, 3 and 6
8	$A(c) \land L(c)$	conjunction, 7 and 4
9	$\exists x\,(A(x) \land L(x))$	existential generalization, 8

We proved an existential claim about the same unknown student.

The archive break-in

A detective knows:

1. Everyone who entered the archive appears in the keycard log.
2. Everyone in the log is an employee.
3. No registered visitor is an employee.
4. Someone in a red coat entered the archive.

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1. Everyone who entered the archive appears in the keycard log.
2. Everyone in the log is an employee.
3. No registered visitor is an employee.
4. Someone in a red coat entered the archive.

Prove: Someone in a red coat was not a registered visitor.

Archive notation

Domain: all people.

$A(x)$	x entered the archive
$K(x)$	x appears in the keycard log
$E(x)$	x is an employee
$V(x)$	x is a registered visitor
$R(x)$	x wore a red coat

Premises

1.;$\forall x\,(A(x) \to K(x))$

2.;$\forall x\,(K(x) \to E(x))$

3.;$\forall x\,(V(x) \to \neg E(x))$

4.;$\exists x\,(A(x) \wedge R(x))$

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3.;$\forall x\,(V(x) \to \neg E(x))$

4.;$\exists x\,(A(x) \wedge R(x))$

Goal

$$\boxed{\exists x\,(R(x) \wedge \neg V(x))}$$

Archive proof (1)

Start with premises 1–4 from the previous frame.

Step	Statement	Reason
5	$A(c) \wedge R(c)$	existential instantiation, 4; c fresh

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7	$R(c)$	simplification, 5

Archive proof (1)

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Step	Statement	Reason
5	$A(c) \land R(c)$	existential instantiation, 4; c fresh
6	$A(c)$	simplification, 5
7	$R(c)$	simplification, 5
8	$A(c) \to K(c)$	universal instantiation, 1
9	$K(c)$	modus ponens, 6 and 8

Archive proof (1)

Start with premises 1–4 from the previous frame.

Step	Statement	Reason
5	$A(c) \wedge R(c)$	existential instantiation, 4; c fresh
6	$A(c)$	simplification, 5
7	$R(c)$	simplification, 5
8	$A(c) \rightarrow K(c)$	universal instantiation, 1
9	$K(c)$	modus ponens, 6 and 8
10	$K(c) \rightarrow E(c)$	universal instantiation, 2
11	$E(c)$	modus ponens, 9 and 10

So the same red-coated person is an employee.

Archive proof (2)

We already know

$$R(c), \qquad E(c).$$

Step	Statement	Reason
12	$V(c) \to \neg E(c)$	universal instantiation, premise 3

Archive proof (2)

We already know

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12	$V(c) \to \neg E(c)$	universal instantiation, premise 3
13	$\neg\neg E(c)$	double negation, $E(c)$

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12	$V(c) \to \neg E(c)$	universal instantiation, premise 3
13	$\neg\neg E(c)$	double negation, $E(c)$
14	$\neg V(c)$	modus tollens, 12 and 13

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12	$V(c) \to \neg E(c)$	universal instantiation, premise 3
13	$\neg\neg E(c)$	double negation, $E(c)$
14	$\neg V(c)$	modus tollens, 12 and 13
15	$R(c) \wedge \neg V(c)$	conjunction, $R(c)$ and 14

Archive proof (2)

We already know

$$R(c), \qquad E(c).$$

Step	Statement	Reason
12	$V(c) \to \neg E(c)$	universal instantiation, premise 3
13	$\neg\neg E(c)$	double negation, $E(c)$
14	$\neg V(c)$	modus tollens, 12 and 13
15	$R(c) \land \neg V(c)$	conjunction, $R(c)$ and 14
16	$\exists x\,(R(x) \land \neg V(x))$	existential generalization, 15

Someone in a red coat was not a registered visitor.

The proof does not identify who that person was.

SECTION 4

Proof methods

Definitions determine the target

For an integer n,

$$n \text{ is even} \iff n = 2k \text{ for some } k \in \mathbb{Z},$$

$$n \text{ is odd} \iff n = 2k + 1 \text{ for some } k \in \mathbb{Z}.$$

To prove that an expression E is even, produce

$$E = 2 \times \underbrace{\text{an integer}}_{\text{required form}}.$$

A definition tells us what the final line must look like.

Direct proof

MUST UNDERSTAND **Direct proof**

Assume the hypotheses and derive the conclusion.

Let m and n be arbitrary odd integers.

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By definition, for some $a, b \in \mathbb{Z}$,

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Direct proof

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Assume the hypotheses and derive the conclusion.

Let m and n be arbitrary odd integers.
By definition, for some $a, b \in \mathbb{Z}$,

$$m = 2a + 1, \qquad n = 2b + 1.$$

Then

$$m + n = 2a + 1 + 2b + 1 = \boxed{2(a + b + 1)}.$$

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Let m and n be arbitrary odd integers.
By definition, for some $a, b \in \mathbb{Z}$,

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Then

$$m + n = 2a + 1 + 2b + 1 = \boxed{2(a + b + 1)}.$$

Since $a + b + 1 \in \mathbb{Z}$, the sum $m + n$ is even.

Three routes for a conditional

To prove $P \to Q$:

Method	Assume	Target
Direct	P	Q

Three routes for a conditional

To prove $P \to Q$:

Method	Assume	Target
Direct	P	Q
Contraposition	$\neg Q$	$\neg P$

Three routes for a conditional

To prove $P \to Q$:

Method	Assume	Target
Direct	P	Q
Contraposition	$\neg Q$	$\neg P$
Contradiction	$P \wedge \neg Q$	an impossibility

Three routes for a conditional

To prove $P \to Q$:

Method	Assume	Target
Direct	P	Q
Contraposition	$\neg Q$	$\neg P$
Contradiction	$P \wedge \neg Q$	an impossibility

Do not mix the starting assumption from one method with the target from another.

Proof by contraposition

MUST UNDERSTAND **Proof by contraposition**

Prove n^2 even $\implies n$ even.

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Prove the contrapositive instead: n odd $\implies n^2$ odd.

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MUST UNDERSTAND Proof by contraposition

Prove n^2 even $\implies n$ even.
Prove the contrapositive instead: n odd $\implies n^2$ odd.
Suppose $n = 2k + 1$ for some $k \in \mathbb{Z}$. Then

$$\begin{aligned} n^2 &= (2k+1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1. \end{aligned}$$

Thus n^2 is odd.

Proof by contraposition

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Suppose $n = 2k + 1$ for some $k \in \mathbb{Z}$. Then

$$\begin{aligned} n^2 &= (2k+1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1. \end{aligned}$$

Thus n^2 is odd.

Therefore, by contraposition, n^2 even implies n even.

Proof by contradiction

MUST UNDERSTAND **Proof by contradiction**

To prove $\sqrt{2}$ irrational, assume the opposite: $\sqrt{2}$ is rational.

Choose integers a, b such that

$$\sqrt{2} = \frac{a}{b}, \qquad b \neq 0,$$

and a/b is in lowest terms.

Proof by contradiction

MUST UNDERSTAND **Proof by contradiction**

To prove $\sqrt{2}$ irrational, assume the opposite: $\sqrt{2}$ is rational.

Choose integers a, b such that

$$\sqrt{2} = \frac{a}{b}, \qquad b \neq 0,$$

and a/b is in lowest terms.
Squaring gives

$$a^2 = 2b^2.$$

Proof by contradiction

MUST UNDERSTAND **Proof by contradiction**

To prove $\sqrt{2}$ irrational, assume the opposite: $\sqrt{2}$ is rational.

Choose integers a, b such that

$$\sqrt{2} = \frac{a}{b}, \qquad b \neq 0,$$

and a/b is in lowest terms.
Squaring gives

$$a^2 = 2b^2.$$

So a^2 is even. The previous theorem implies that a is even.
Write $a = 2k$ for some $k \in \mathbb{Z}$.

The contradiction in the $\sqrt{2}$ proof

Substitute $a = 2k$ into $a^2 = 2b^2$:

$$\begin{aligned} (2k)^2 &= 2b^2, \\ b^2 &= 2k^2. \end{aligned}$$

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This contradicts the choice of a/b in lowest terms. Therefore $\sqrt{2}$ is irrational.

Biconditional proofs

$$P \leftrightarrow Q \quad \iff \quad (P \to Q) \wedge (Q \to P).$$

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Forward: direct

$n = 2k \Rightarrow n^2 = 2(2k^2)$.

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Odd n has odd square.

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Odd n has odd square.

$$\begin{tikzcd} P_1 \arrow[r] & P_2 \arrow[r] & P_3 \arrow[ll, bend left] \end{tikzcd}$$

For several statements, a cycle lets every statement reach every other.

SECTION 5

Other proof forms

Proof by cases

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Let n be an arbitrary integer. Every integer is **even or odd**, so these two cases cover the domain.

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Case 2: n is odd

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If $n = 2k+1$, then $n+1 = 2(k+1)$, so

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Both cases produce 2 times an integer; therefore $n(n+1)$ is even for every integer n.

Existence and uniqueness

Existence: $\exists x\, P(x)$

Give one object and verify it.

$$64 = 8^2 = 4^3$$

Exactly one: $\exists! x\, P(x)$

Prove both:

1. at least one object works;
2. any two working objects are equal.

One example does not prove uniqueness; “at most one” does not prove existence.

A uniqueness proof

Assume $a, b, c \in \mathbb{R}$ and $a \neq 0$.

Existence

$$x = \frac{c - b}{a}$$

Uniqueness

A uniqueness proof

Assume $a, b, c \in \mathbb{R}$ and $a \neq 0$.

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If x_1, x_2 both work, then

$$ax_1 + b = ax_2 + b.$$

Hence $a(x_1 - x_2) = 0$. Since $a \neq 0$, $x_1 = x_2$.

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The condition $a \neq 0$ justifies every division and the uniqueness step.

Counterexamples and proof errors

Refute a universal claim

Find one allowed a for which $P(a)$ is false.

"Every prime is odd" is false:

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- Was the target assumed circularly?

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One counterexample refutes a universal claim; many unexhausted examples do not prove one.

SECTION 6

Proof strategy

Choosing a proof method

Claim	Natural first move
$\forall x\, P(x)$	choose an arbitrary x
$\exists x\, P(x)$	find or derive one example
$P \to Q$	assume P (direct), $\neg Q$ (contraposition), or $P \wedge \neg Q$ (contradiction)
$P \leftrightarrow Q$	prove both directions
$\exists! x\, P(x)$	separate existence and uniqueness
universal claim suspected false	search for a counterexample

The symbols tell us what evidence the proof must deliver.

Method-selection practice

Choose a promising first method. State the opening assumption or witness.

1. If n^2 is odd, then n is odd.
2. There exists an even prime.
3. For every integer n, the number $n^2 + n$ is even.
4. If $a \neq 0$, then $ax + b = c$ has exactly one real solution.

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1	contraposition: assume n even
2	witness: 2
3	even/odd cases, or consecutive parity
4	separate existence from uniqueness

Proof checklist

1. State the **domain, assumptions, and target**.
2. Expand the definitions that determine the required form.
3. Work forward; use backward reasoning to discover what would suffice.
4. Check implication directions, exhausted cases, side conditions, and the final target.

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Next: Sets and Functions.
We will use proofs to establish identities, inclusions, and properties of mappings.